14 Rachel has 10 sweets in a bag.

7 of the sweets are yellow.
3 of the sweets are red.

Rachel takes random sweets from the bag, one at a time, until she gets a red sweet.
She does not replace the sweets that she takes from the bag.

Rachel stops as soon as she gets a red sweet.

Work out the probability that Rachel takes no more than three sweets from the bag.

No more than 3 means the first red occurs

on draw 1, 2 or 3.

Use the complement: first three are all yellow.

$P(YYY) = \frac{7}{10} \times \frac{6}{9} \times \frac{5}{8}$

1 mark
Without replacement.

$= \frac{210}{720} = \frac{7}{24}$

1 mark
Subtract from 1.

$P(\text{no more than 3}) = 1 - \frac{7}{24} = \frac{17}{24}$

1 mark
17/24.

.....

(Total for Question is 14 is 3 marks)